

CONTENT

ACKNOWLEDGMENTS	2
LIST OF ABBREVIATIONS	3
ABSTRACT	4
I. INTRODUCTION	5
II. MATERIALS AND METHODS	16
2.1 Computing hardware	16
2.2 Software: data analysis and visualisation tools	17
2.3 Dataset	17
2.3.1 Preliminary preparation of the dataset for the present study	17
2.3.2 Dataset structure	19
2.3.3 Additional explanations	22
III. RESULTS	24
3.1 Analysis workflow and files management	24
3.2 Preliminary analysis: dataset structure	26
3.3 Cluster combinations count in the initial dataset	27
3.4 Standardizing the table form. Switched dataset	30
3.5 Cluster count in the switched dataset	32
3.6 Total assemblage of cluster combinations: pattern dataset. Unique patterns	34
3.7 Gene pairs lists and single gene lists	37
3.8 Difference between single gene lists	39
IV. DISCUSSION	42
4.1 Results interpretation	42
4.1.1. Preliminary analysis of the initial dataset	42
4.1.2. Cluster combinations count	43
4.1.3 Patterns	45
4.1.4 Gene lists	46
4.2 Consequences and implications	47
V. CONCLUSION	50
5.1 Key findings	50
5.2 Final thoughts	50
5.3 Recommendations	50
5.4 Final words	51
VI. FUTURE WORK AND LIMITATIONS	52
6.1 Future research	52
6.2 Limitations	52
REFERENCES	54
APPENDICIES	61